

# AIM2- Preparations

## Day 1: Python Fundamentals

### 1 Variables

Let's start with something you already know from Java.

#### Java

You might write:

```
int age = 25;  
String name = "Krishna";  
double score = 88.5;  
boolean passed = true;
```

Python is much simpler:

```
age = 25  
name = "Krishna"  
score = 88.5  
passed = True
```

Notice something?

#### Python doesn't require:

```
int  
String  
double  
boolean
```

**before the variable.**

*Python figures out the type automatically.*

This is called **dynamic typing**.

## 2 Python Data Types

There are several important built-in types.

For today, focus on these:

| Type     | Example   | Meaning        |
|----------|-----------|----------------|
| int      | 25        | Whole number   |
| float    | 88.5      | Decimal number |
| str      | "Krishna" | Text           |
| bool     | True      | True/False     |
| NoneType | None      | No value       |

### Let's use type()

Python has a built-in function:

**type()**

**It tells you what type something is.**

Try:

```
name = "Krishna"
age = 25
score = 88.5
is_student = True

print(type(name))
print(type(age))
print(type(score))
print(type(is_student))
```

You'll get approximately:

```
<class 'str'>
<class 'int'>
<class 'float'>
<class 'bool'>
```

This is something you'll use frequently when working with data.

## 3 Strings — str

Strings are text.

```
name = "Krishna"
city = "London"
program = "AIM2"
```

You can use either:

"Hello"

or:

'Hello'

Both are strings.

### String concatenation

You can combine strings:

```
first_name = "Krishna"
last_name = "Pandya"

full_name = first_name + " " + last_name

print(full_name)
```

Output:

Krishna Pandya

This is similar to Java's:

```
String fullName = firstName + " " + lastName;
```

## ★ 4 f-strings — VERY IMPORTANT

Python has a fantastic way of putting variables inside strings.

It's called an **f-string**.

```
name = "Krishna"
age = 25

print(f"My name is {name} and I am {age} years old.")
```

Output:

*My name is Krishna and I am 25 years old.*

The important syntax is:

```
f"... {variable} ..."
```

You'll use this **all the time** in Python.

For example:

```
score = 88.5

print(f"My score is {score}")
```

## 5 String Methods

Python strings have useful methods.

Try:

```
text = "Artificial Intelligence"

print(text.upper())
print(text.lower())
print(text.title())
```

You'll get:

```
ARTIFICIAL INTELLIGENCE
artificial intelligence
Artificial Intelligence
```

You can also find the length:

```
print(len(text))
```

## More useful methods

```
text = "  Python is Amazing  "

print(text.strip())
print(text.lower())
print(text.upper())
```

`strip()` removes extra spaces at the beginning and end.

Another important one:

```
text = "I love Java"

print(text.replace("Java", "Python"))
```

Result:

I love Python

## Numbers

Python supports normal mathematical operations.

```
a = 20
b = 6

print(a + b)
print(a - b)
print(a * b)
print(a / b)
```

Result:

```
26
14
120
3.3333333333333335
```

But there are some operators you need to know.

### **/ Division**

```
10 / 3
```

Result:

```
3.3333333333333335
```

### **// Floor division**

```
10 // 3
```

Result:

```
3
```

## % Modulus

```
10 % 3
```

Result:

```
1
```

## \*\* Power

```
2 ** 3
```

Result:

```
8
```

So remember:

```
/ → division
```

```
// → floor division
```

```
% → remainder
```

```
** → power
```

These are important for Python.

## 7 Boolean

Boolean has only two values:

True

False

Notice the capital letters.

Correct:

```
is_student = True
```

**Incorrect:**

```
is_student = true
```

**Python is case-sensitive.**

For example:

```
age = 25

print(age > 18)
print(age < 18)
print(age == 25)
```

Output:

```
True
False
True
```

We'll use booleans heavily tomorrow/Day 3 when we start conditions.

## None

This is an important Python concept.

```
result = None
```

It means:

**There is currently no value.**

For example:

```
student_email = None
```

Maybe the student hasn't provided an email yet.



None is roughly similar to null in Java.

## Java

```
String email = null;
```

## Python

```
email = None
```

Remember:

Java → null

Python → None

# input() — Getting information from the user

Now we're going to make our program interactive.

Try:

```
name = input("Enter your name: ")  
  
print(f"Hello {name}!")
```

When you run it:

```
Enter your name: Krishna  
Hello Krishna!
```

Pretty simple.

But there's an **important Python detail**.

## 10 input() always returns a string

Suppose we do:

```
age = input("Enter your age: ")
```

Even if you type:

25

Python receives:

"25"

It's a **string**, not an integer.

You can check:

```
age = input("Enter your age: ")
```

```
print(type(age))
```

You'll get:

```
<class 'str'>
```

## 11 Type Conversion

So if we need an integer:

```
age = int(input("Enter your age: "))
```

Now:

```
print(type(age))
```

gives:

```
<class 'int'>
```

Similarly:

```
score = float(input("Enter your score: "))
```

creates a float.

You should know these:

```
int()  
float()  
str()  
bool()
```

For example:

```
number = "25"
```

```
number = int(number)
```

```
print(number)
```

```
print(type(number))
```

## Your First Exercise

Now I want **you** to write something.

Don't copy a solution from me because this is where your learning starts.

Create a program that asks the user for:

Name

Age

Program

Math score

Python score

AI score

Then calculate:

Total

Average

And print something like:

```
=====
Student Performance
=====
```

Name: Krishna

Age: 25

Program: AIM2

Math: 85

Python: 92

AI: 88

Total: 265

Average: 88.33

```
=====
```